

1 Assignment 4a — Neutrino Oscillations in Matter



The neutrino time evolution in matter is governed by Schrödinger equation with an effective Hamiltonian that includes both the vacuum term and a matter potential term.

In class, we saw that we can write the time evolution as

$$i \frac{d}{dt} \begin{pmatrix} \nu_e \\ \nu_\mu \end{pmatrix} = \left[U \begin{pmatrix} E_1^0 & 0 \\ 0 & E_2^0 \end{pmatrix} U^\dagger + \begin{pmatrix} V_{CC} & 0 \\ 0 & 0 \end{pmatrix} \right] \begin{pmatrix} \nu_e \\ \nu_\mu \end{pmatrix}, \quad (1.1)$$

where $E_i^0 \simeq p + \frac{m_i^2}{2p}$ are the energy eigenvalues in vacuum, $U = R(\theta) \equiv \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$, and V_{CC} is the charged-current matter potential.

- (4 points) 1. Define the effective mixing angle in matter, θ_m , such that the effective Hamiltonian can be diagonalized as

$$\hat{H}_m = U_m \begin{pmatrix} E_1^m & 0 \\ 0 & E_2^m \end{pmatrix} U_m^\dagger, \quad \text{where} \quad i \frac{d}{dt} \nu_i^m = E_i^m \nu_i^m, \quad (1.2)$$

where $U_m = R(\theta_m)$ is the mixing matrix with angle θ_m that takes us to the diagonal basis in matter $\nu_{\text{flavor}} = U_m \nu^m$. The script m indicates that these are the matter eigenstates and eigenvalues. We have implicitly assumed that the matter potential is constant throughout the neutrino propagation (take $t \simeq x$ and Eq. (1.1) above tells you how the state evolves as it propagates from source to detector). Derive an expression for the eigenvalues E_i^m and $\sin 2\theta_m$ in terms of the vacuum mixing angle θ , the neutrino energy E_ν , the mass-squared difference $\Delta m^2 = m_2^2 - m_1^2$, and the matter potential V_{CC} . Define a matter-basis mass splitting as $\Delta m_m^2 / (2E_\nu) \equiv E_2^m - E_1^m$ and write your expression for Δm_m^2 as a function of these parameters.

(Hint: Recall that diagonal terms in the time evolution will not contribute to oscillations.)

(Hint 2: Recall also that a real symmetric 2×2 matrix $\begin{pmatrix} a & c \\ c & b \end{pmatrix}$ can be diagonalized by an orthogonal transformation with angle θ such that $\tan 2\theta = \frac{2c}{b-a}$ or $\sin 2\theta = \frac{2c}{\sqrt{(b-a)^2 + 4c^2}}$.

(Hint 3: Your answer should look like $\sin^2 2\theta_m = \sin^2 2\theta / R$ and $\Delta m_m^2 = \Delta m^2 \sqrt{R}$ for some R factor.)

- (2 points) 2. Using the expression derived in Question 1, find the condition for which the effective mixing angle in matter θ_m becomes maximal (i.e., $\theta_m = \pi/4$). What does this mean for oscillations? This condition is known as the Mikheyev-Smirnov-Wolfenstein (MSW) resonance condition.
- (2 points) 3. Given that $V_{CC} = \sqrt{2}G_F N_e$, where G_F is the Fermi constant and N_e is the electron number density in matter, estimate the neutrino energy E_ν at which the MSW reso-

nance occurs for solar neutrinos propagating through the Sun

(Hint: take $N_e \sim 100N_A \text{ cm}^{-3}$, where N_A is Avogadro's number. Assume $\Delta m^2 = 7.5 \times 10^{-5} \text{ eV}^2$ and $\theta = 33^\circ$.)

- (**2 points**) 4. The survival probability in matter becomes $P(\nu_e \rightarrow \nu_e) = 1 - \sin^2 2\theta_m \sin^2 \left(\frac{\Delta m_m^2 t}{4E_\nu} \right)$, analogous to the vacuum oscillation probability. What happens when the MSW resonance condition is exactly satisfied throughout the entire neutrino propagation for $\Delta m^2 t / 2E_\nu \gg 1$? What about when $\Delta m_M^2 t / 2E_\nu \ll 1$ (expand the sine functions)?

- (**Bonus 2 points**) 5. For a changing matter density, the effective mixing angle θ_m also changes along the neutrino trajectory. Derive the adiabaticity condition for which a neutrino that starts as a matter eigenstate ν_1^m (or ν_2^m) remains in that eigenstate throughout the propagation. What happens when this condition is satisfied?



Solutions and discussion for Assignment 4b.

Question 1. We start from the effective Hamiltonian in matter,

$$\hat{H}_m = U \begin{pmatrix} E_1^0 & 0 \\ 0 & E_2^0 \end{pmatrix} U^\dagger + \begin{pmatrix} V_{CC} & 0 \\ 0 & 0 \end{pmatrix}. \quad (1.3)$$

As mentioned in the hint, we can drop any term proportional to the identity matrix since it will not contribute to oscillations. Thus, we can rewrite the vacuum term as

$$U \begin{pmatrix} E_1^0 & 0 \\ 0 & E_2^0 \end{pmatrix} U^\dagger = \frac{E_1^0 + E_2^0}{2} \mathbb{1} + \frac{\Delta m^2}{4E_\nu} \begin{pmatrix} -\cos 2\theta & \sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{pmatrix}, \quad (1.4)$$

where we used $E_i^0 \simeq p + \frac{m_i^2}{2p}$ and dropped the p term.